Areej Najd
For cosmetics



MEMORY BOOST

Citicoline & Omega-3 Syrup · 120 ml

Complete brain nutrition — neuronal membranes, neurotransmitters, energy and homocysteine control — in one easy-to-take syrup.

13 active ingredients · Sugar-free sweeteners · No fishy taste · Food supplement

250 mg
DHA
Contributes to the maintenance of normal brain function

250 mg
Citicoline (CDP-choline)
Choline + cytidine donor for phosphatidylcholine and acetylcholine

50 mg
L-Theanine
Relaxed alertness without sedation

SCIENTIFIC RATIONALE

- **Membrane building blocks**
  DHA is the main omega-3 of neuronal membranes; citicoline supplies choline + cytidine for phosphatidylcholine.
- **Omega-3 + B-vitamins synergy**
  B6, folate and B12 lower homocysteine; their brain benefit depends on adequate omega-3 status.[8]
- **Acetylcholine support**
  Two choline sources feed acetylcholine, the key neurotransmitter of memory and learning.
- **Iron, zinc, tyrosine & PS**
  Iron and zinc contribute to normal cognitive function; tyrosine and PS complete the profile.

CLINICAL EVIDENCE ON THE INGREDIENTS — full study details on pages 2–3

MIDAS trial[1]
Yurko-Mauro, 2010
485 healthy older adults with age-related cognitive decline · DHA 900 mg/day vs placebo, 24 weeks → significant improvement in **learning and episodic memory**.

Citicoline & attention[2]
McGlade, 2012
60 healthy women (40–60 y) · citicoline **250 mg** or 500 mg/day vs placebo, 28 days → **fewer attention errors** in the 250 mg group.

Citicoline & memory[3]
Nakazaki, 2021
100 healthy older adults with age-associated memory impairment · citicoline 500 mg/day, 12 weeks → improved **overall and episodic memory** vs placebo (RCT).

L-Theanine & alpha waves[5]
Nobre, 2008
L-theanine **50 mg** increased **alpha-wave activity** on EEG vs placebo — a marker of relaxed, attentive mental state.

COMPOSITION PER 10 ML SERVING

Fish oil (omega-3)	**1100 mg**	Phosphatidylserine	**50 mg**	Vitamin B6	**1.5 mg**
EPA	**350 mg**	N-acetyl-L-tyrosine	**50 mg**	Folic acid	**200 mcg**
DHA	**250 mg**	L-Theanine	**50 mg**	Vitamin B12	**6 mcg**
Citicoline	**250 mg**	Iron	**7 mg**		
Choline bitartrate	**100 mg**	Zinc	**5 mg**		

SUITABLE FOR

Students & adolescents (12+) during study and exams · Adults with high mental workload or mental fatigue · Older adults with age-related memory decline · Low dietary intake of fish / omega-3

Full references and study details: see pages 2–3.

SFDA Reg. FOM-24-272344-FOP-251125-1028812-C-12688-881856 · Dr. Organic

Memory Boost — clinical evidence in detail

Published studies on the key ingredients of Memory Boost: full citation, study design, participants, the amount studied, duration and main findings.

1 · MIDAS trial — DHA and memory

Omega-3 (DHA)

Yurko-Mauro K, McCarthy D, Rom D, et al. *Beneficial effects of docosahexaenoic acid on cognition in age-related cognitive decline.* Alzheimer's & Dementia. 2010;6(6):456–464 · doi:10.1016/j.jalz.2010.01.013

Healthy adults aged ≥55 with age-related cognitive decline. The DHA group made **significantly fewer errors on a visuospatial learning and episodic memory test** (CANTAB PAL) than placebo, and DHA was well tolerated.

Memory Boost: DHA 250 mg + EPA 350 mg

Design	**RCT, double-blind**
Participants	**485 adults ≥55 y**
Intake	**Algal DHA 900 mg/day**
Duration	**24 weeks**

2 · Citicoline and attention in healthy women

Citicoline

McGlade E, Locatelli A, Hardy J, et al. *Improved attentional performance following citicoline administration in healthy adult women.* Food and Nutrition Sciences. 2012;3(6):769–773 · doi:10.4236/fns.2012.36103

Healthy women aged 40–60. Compared with placebo, the citicoline groups showed **better attentional performance** on a continuous performance test, with **fewer errors in the 250 mg group**.

Same citicoline amount as Memory Boost: 250 mg

Design	**RCT, double-blind**
Participants	**60 women, 40–60 y**
Intake	**Citicoline 250 or 500 mg/day**
Duration	**28 days**

3 · Citicoline and memory in older adults

Citicoline

Nakazaki E, Mah E, Sanoshy K, Citrolo D, Watanabe F. *Citicoline and memory function in healthy older adults: a randomized, double-blind, placebo-controlled clinical trial.* The Journal of Nutrition. 2021;151(8):2153–2160 · doi:10.1093/jn/nxab119

Healthy adults aged 50–85 with age-associated memory impairment. Citicoline **improved overall memory, especially episodic memory**, compared with placebo.

Memory Boost: citicoline 250 mg

Design	**RCT, double-blind**
Participants	**100 adults, 50–85 y**
Intake	**Citicoline 500 mg/day**
Duration	**12 weeks**

4 · Cochrane review — citicoline

Citicoline

Fioravanti M, Yanagi M. *Cytidinediphosphocholine (CDP-choline) for cognitive and behavioural disturbances associated with chronic cerebral disorders in the elderly.* Cochrane Database of Systematic Reviews. 2005;(2):CD000269 · doi:10.1002/14651858.CD000269.pub3

Systematic review of randomised trials in older adults with chronic cerebral disorders. Found **a positive effect of citicoline on memory and behaviour** in the short to medium term.

Supports citicoline's cognitive role

Design	**Systematic review**
Participants	**Older adults**
Intake	**Various doses**
Duration	**Short–medium term**

5 · L-Theanine and alpha brain waves

L-Theanine

Nobre AC, Rao A, Owen GN. *L-theanine, a natural constituent in tea, and its effect on mental state.* Asia Pacific Journal of Clinical Nutrition. 2008;17(S1):167–168

Healthy young adults, EEG recording after a single dose. L-theanine **increased alpha-wave activity** compared with placebo — a marker of a relaxed but alert mental state.

Same L-theanine amount as Memory Boost: 50 mg

Design	**Placebo-controlled**
Participants	**Healthy young adults**
Intake	**L-theanine 50 mg**
Duration	**Single dose**

Memory Boost — clinical evidence in detail (continued)

6 · L-Theanine, stress and cognition

L-Theanine

Hidese S, Ogawa S, Ota M, et al. *Effects of L-theanine administration on stress-related symptoms and cognitive functions in healthy adults: a randomized controlled trial.* Nutrients. 2019;11(10):2362 · doi:10.3390/nu11102362

Healthy adults. L-theanine **reduced stress-related symptoms**, improved sleep quality and some cognitive functions (verbal fluency, executive function) compared with placebo.

Memory Boost: L-theanine 50 mg

Design	**RCT, double-blind, crossover**
Participants	**30 healthy adults**
Intake	**L-theanine 200 mg/day**
Duration	**4 weeks**

7 · VITACOG — B vitamins and brain atrophy

Folate · B6 · B12

Smith AD, Smith SM, de Jager CA, et al. *Homocysteine-lowering by B vitamins slows the rate of accelerated brain atrophy in mild cognitive impairment: a randomized controlled trial.* PLoS One. 2010;5(9):e12244 · doi:10.1371/journal.pone.0012244

Older adults with mild cognitive impairment (MRI sub-study). B vitamins **slowed the rate of brain atrophy by about 30%**, with a larger effect in those with high homocysteine.

Memory Boost: folate + B6 + B12

Design	**RCT, double-blind**
Participants	**168 adults ≥70 y (MRI)**
Intake	**Folic acid 0.8 mg + B6 20 mg + B12 0.5 mg**
Duration	**24 months**

8 · VITACOG — omega-3 status and brain atrophy

Omega-3 + B vitamins

Jernerén F, Elshorbagy AK, Oulhaj A, et al. *Brain atrophy in cognitively impaired elderly: the importance of long-chain ω-3 fatty acids and B vitamin status in a randomized controlled trial.* Am J Clin Nutr. 2015;102(1):215–221 · doi:10.3945/ajcn.114.103283

Analysis of the VITACOG trial. B vitamins **slowed brain atrophy only in participants with good omega-3 (EPA + DHA) status**; with low omega-3 there was no clear benefit.

Rationale for omega-3 + B vitamins together

Design	**RCT secondary analysis**
Participants	**VITACOG MRI cohort**
Intake	**B vitamins as in VITACOG**
Duration	**24 months**

9 · VITACOG — omega-3 status and cognition

Omega-3 + B vitamins

Oulhaj A, Jernerén F, Refsum H, Smith AD, de Jager CA. *Omega-3 fatty acid status enhances the prevention of cognitive decline by B vitamins in mild cognitive impairment.* J Alzheimers Dis. 2016;50(2):547–557 · doi:10.3233/JAD-150777

Analysis of the VITACOG trial. B vitamins **slowed cognitive decline only when omega-3 status was good**.

Rationale for omega-3 + B vitamins together

Design	**RCT secondary analysis**
Participants	**VITACOG cohort, MCI**
Intake	**B vitamins as in VITACOG**
Duration	**24 months**

The studies above evaluated the individual ingredients (or combinations of them), not the finished Memory Boost product. Amounts used in some studies differ from those in Memory Boost; the relevant amount in the product is shown under each study. Memory Boost is a food supplement and is not intended to diagnose, treat or prevent any disease.